

Pengyu Zhang

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📍 Amsterdam, The Netherlands  Fourth-year Ph.D. Candidate

Education

- 2022 – 2026** **Ph.D. in Computer Science**, Informatics Institute, University of Amsterdam (UvA), The Netherlands.
Supervisors: Prof. Dr. Paul Groth, Dr. Klim Zaporozhets.
- 2019 – 2022** **M.Eng. in Control Engineering**, Beijing University of Technology (BJUT), China.
Supervisor: Prof. Dr. Yong Zhang.

Research Interests

I build models that stay accurate as data evolves over time and across modalities. My work tackles temporal drift in entity linking and knowledge graphs, fuses graph structure, text, images, and time into robust representations, and uses vision-language models and LLMs to enrich data at scale, turning these advances into measurable gains for retrieval, link prediction, and recommendation.

Keywords: temporal entity linking, multimodal knowledge graphs, graph contrastive learning, vision-language models, LLM-based enrichment, time-aware recommendation.

Research Experience

- TimeRoute** **Lifted multi-modal recommendation accuracy by up to 6%** on TikTok, Amazon-Baby, and Amazon-Sports over strong baselines.
Designed a time-aware modality-routing model with split-frequency sinusoidal timestamp embeddings and a Hybrid Mixture-of-Experts (long-term, short-term, adaptive), then conditioned a diffusion-based graph reconstructor for denoising and completion — solving the failure of static fusion to model modalities that matter on different time scales.
- Time Imprint** **Improved Hits@1 by up to 4.81% overall and up to 200%** on the hardest, most confusable entity pairs.
Modeled time as a third modality with sinusoidal temporal encoding, attention pooling over selected timestamps, cross-modal Transformer fusion, and three-view contrastive alignment, so entities remain distinguishable even when their text and images look alike.
- Beyond Images** **Boosted Hits@1 by up to +7% overall and +333%** on logo/symbol subsets where visuals are ambiguous.
Built an automated data-enrichment pipeline that retrieves additional web images, converts images to text via vision-language models, and fuses the descriptions with an LLM into entity-aligned summaries — directly addressing limited image coverage in multi-modal knowledge graphs.
- CYCLE** **Beat prior state-of-the-art by 13.9% (1-year gap) and 17.8% (3-year gap)**, with the largest gains on low-degree entities.
Proposed graph contrastive learning across yearly snapshots — newly added relationships as positives, removed ones as negatives — and released the GCL-TempEL benchmark built on Wikidata5M, tackling the degradation of entity linkers as knowledge graphs evolve.
- TIGER** **Outperformed SpEL by +16.24% (1-year gap) and +20.93% (3-year gap)**, with notable gains on high-degree entities.
Combined distinct/shared graph-convolution modules for structure and feature graphs with text bi-encoder representations, and introduced the Graph-TempEL benchmark to measure temporal robustness of entity linking.

Publications

- 2026** **TimeRoute: Time-Aware Modality Routing and Diffusion for Multi-Modal Recommendation.**
Pengyu Zhang, Yangqin Jiang, Klim Zaporozets, Congfeng Cao, and Paul Groth.
Preprint.
- 2026** **Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs.**
Pengyu Zhang, Klim Zaporozets, Congfeng Cao, Jia-Hong Huang, and Paul Groth.
Preprint.
- 2026** **Beyond Images: Are a Thousand Words Better Than a Single Picture? A Framework for Multi-Modal Knowledge Graph Dataset Enrichment.**
Pengyu Zhang, Klim Zaporozets, Congfeng Cao, Jie Liu, and Paul Groth.
In *The European Semantic Web Conference (ESWC 2026)*.
 Paper  Code
- 2024** **CYCLE: Cross-Year Contrastive Learning in Entity-Linking.**
Pengyu Zhang, Congfeng Cao, Klim Zaporozets, and Paul Groth.
In *International ACM Conference on Knowledge and Information Management (CIKM 2024)*.
 Paper  Code
- 2024** **TIGER: Temporally Improved Graph Entity Linker.**
Pengyu Zhang, Congfeng Cao, and Paul Groth.
In *European Conference on Artificial Intelligence (ECAI 2024)*.
 Paper  Code

Selected Talks

- 2026** Conference talk, *European Semantic Web Conference (ESWC 2026)*, “Beyond Images.”
- 2024** Conference talk, *ACM Conference on Information and Knowledge Management (CIKM 2024)*, “CYCLE.”
- 2024** Conference talk, *European Conference on Artificial Intelligence (ECAI 2024)*, “TIGER.”

Skills

- Research** Multimodal knowledge graphs, entity linking, link prediction, temporal graph learning, graph neural networks, and knowledge acquisition.
- Technical** Python, PyTorch, large language models, vision-language models, graph representation learning pipelines, and benchmark dataset construction.
- Languages** Chinese (native); English (fluent, IELTS 7.0).